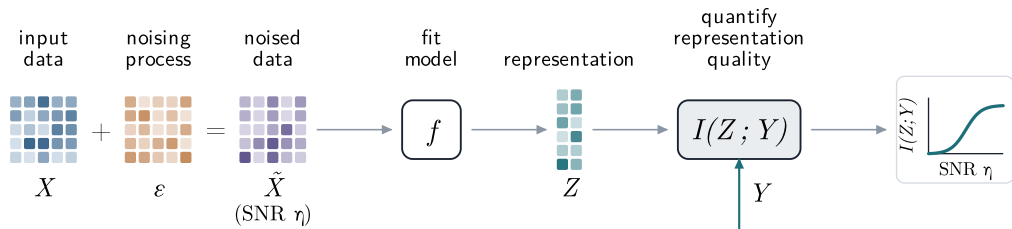


aauxiliary signal Y 
Spatial neighbors
MERFISH

Clonal neighbors
LARRY

Surface protein
CITE-seq

Developmental time
time-series scSeq
b

Data modality	Tag	Auxiliary signal	N samples	Mean UMI / cell	Model families	Noising process
Single-cell & spatial transcriptomics	Spatial	Spatial neighbor expression (MERFISH)	6×10^4	3.7×10^2	scVI Geneformer STATE PCA Random projection	Transcript count downsampling
	Clonal	Clonal neighbor expression (LARRY)	1×10^5	2.6×10^3		
	Protein	Protein abundance (CITE-seq)	1×10^5	8.1×10^3		
	Temporal	Embryo stage	1×10^7	2.5×10^3		
Images	TissueMNIST	Kidney cell type (fluorescence microscopy)	2.4×10^5	—	MobileNetV2	Pixel-wise additive Gaussian; pixelation
	Caltech101	Image label	2.7×10^3	—		
Protein sequences	Spike	Collection month (SARS-CoV-2 spike)	6.3×10^4	—	ESM2	Amino-acid substitution error